

Disproof of conjecture 00000009629: the full 2-shift and the full 4-shift have order-isomorphic dimension groups but are not orbit equivalent

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: Orbit equivalence is an orbit-preserving Borel isomorphism; Ornes theory connects dimension groups with orbit equivalence classification. Conjecture: Two mixing SFTs are orbit equivalent if and only if their dimension groups are order-isomorphic; entropy takes all nonnegative values within an orbit equivalence class, and entropy-changing orbit equivalences are realizable by explicit Kakutani-Rokhlin tower reconstructions. (orbit equivalence dimension criterion)

The conjecture is a conjunction of three clauses. We refute the first clause, and in fact its “if” direction: we exhibit two mixing SFTs whose dimension groups are order-isomorphic but which are not orbit equivalent. This refutes the conjunction, whatever the second and third clauses mean. We do not address the entropy clause or the Kakutani-Rokhlin clause.

Reading.

- *Orbit equivalence* is exactly as in the Definition line: a bijection $\varphi : X \rightarrow Y$, Borel measurable in both directions (Borel σ -algebras of the subspace topologies), with $\varphi(\text{orb}(x)) = \text{orb}(\varphi(x))$ for every x , where $\text{orb}(x) = \{\sigma^k x : k \in \mathbb{Z}\}$. Our non-equivalence argument uses only that φ is a bijection with $\varphi(\text{orb}(x)) \subseteq \text{orb}(\varphi(x))$, so it also covers topological orbit equivalence and every weaker orbit-preserving notion that still consists of a bijection of the entire spaces mapping each orbit into an orbit (it does not cover notions that discard exceptional sets or allow non-bijective maps).
- *SFTs* are presented as edge shifts X_A of square matrices A over $\mathbb{Z}_{\geq 0}$; every SFT is conjugate to an edge shift [W1].
- *Mixing* is topological mixing of (X, σ) [W2].
- *Dimension group* is the dimension group of A with its positive cone, $(\mathcal{G}_A, \mathcal{G}_A^+)$ [S]; *order-isomorphic* means a group isomorphism carrying $\mathcal{G}_A^+$ onto $\mathcal{G}_B^+$. The text says “dimension groups are order-isomorphic” (Chinese: “wei shu qun you xu tong gou”, ordered isomorphism of the dimension groups). The literature separates the dimension group with its order from the *dimension triple*, which adds the shift automorphism δ_A ([S], and [E]: “One can further endow this group by an order and the automorphism induced by the shift map to obtain the so called dimension triple”).

Readings not refuted. (a) The reading in which “dimension groups” means the dimension triple $(\mathcal{G}_A, \mathcal{G}_A^+, \delta_A)$, i.e. order isomorphisms that also intertwine the shift automorphisms. Under it, [2] and [4] are not equivalent and our witness does not apply. (b) The reading in which “dimension group” means the Giordano-Putnam-Skau group $K^0(X, \sigma) = C(X, \mathbb{Z})/(1 - \sigma)C(X, \mathbb{Z})$ rather than Krieger’s dimension group.

2 Definitions (as formalized)

Definition 1 (Edge shift [E]). Let A be an $r \times r$ matrix over $\mathbb{Z}_{\geq 0}$. Its edges are the triples (i, j, k) with $k < A_{ij}$, from i to j (the canonical labelling of [E]). The edge shift X_A is the set of bi-infinite edge sequences $(e_k)_{k \in \mathbb{Z}}$ with $i(e_{k+1}) = t(e_k)$ for all k . With the discrete topology on edges, X_A is closed and shift-invariant in $E^{\mathbb{Z}}$; in Lean it is a Mathlib `Subshift` (`edgeShift`). The shift is $(\sigma x)_i = x_{i+1}$ (Mathlib’s `shift 1`).

Definition 2 (SFT, mixing). `IsSFT X`: X is Mathlib’s `forbidden F` for a finite set F of patterns. `IsMixing X`: for all nonempty open $U, V \subseteq X$ there is N with $\sigma^n(U) \cap V \neq \emptyset$ for every $n > N$ [W2].

Definition 3 (Dimension group [S, Sec. 2]). Let A be $r \times r$ over $\mathbb{Z}_{\geq 0}$, acting on row vectors. $R(A) = \mathbb{Q}^r A^r$ is the eventual range,

$$\mathcal{G}_A = \{x \in R(A) : xA^k \in \mathbb{Z}^r \text{ for some } k \geq 0\}, \quad \mathcal{G}_A^+ = \{x \in R(A) : xA^k \in \mathbb{Z}_{\geq 0}^r \text{ for some } k \geq 0\}.$$

In Lean, `dimGroup A` is an `AddSubgroup` of $\mathbb{Q}^r$ (closure under $+$ and $-$ is proved) and `dimCone A` is $\mathcal{G}_A^+$. `DimOrderIso A B` asserts an additive group isomorphism $e : \mathcal{G}_A \rightarrow \mathcal{G}_B$ with $x \in \mathcal{G}_A^+ \iff e(x) \in \mathcal{G}_B^+$.

Definition 4 (Orbit equivalence). `OrbitEquivalent X Y`: there is a bijection $\varphi : X \rightarrow Y$, measurable from `borel X` to `borel Y` with measurable inverse, such that $\varphi(\text{orb}(x)) = \text{orb}(\varphi(x))$ for all $x \in X$.

The witnesses are $X_{[2]}$ and $X_{[4]}$ for the 1×1 matrices $[n]$: one vertex with n loops. Since every edge is a loop, every edge sequence is allowed, so $X_{[n]}$ is the full shift on an n -letter alphabet (Lean: `carrier_full, card_edge_full`).

3 Proof

Lemma 5 (SFT and mixing). $X_{[n]}$ equals `forbidden  $\emptyset$` , so it is an SFT, and it is topologically mixing.

Proof. Its carrier is all of $E^{\mathbb{Z}}$. For mixing, let U, V be nonempty open in $E^{\mathbb{Z}}$, $u \in U$, $v \in V$. By the definition of the product topology there are finite $I, J \subset \mathbb{Z}$ with $\{y : y|_I = u|_I\} \subseteq U$ and $\{y : y|_J = v|_J\} \subseteq V$. Let $M = \sum_{i \in I \cup J} |i|$. For $n > 2M$ put $x_i = u_i$ for $i \in I$ and $x_i = v_{i-n}$ otherwise. Then $x \in U$, and for $j \in J$ we have $n + j > M \geq \max I$, so $(\sigma^n x)_j = x_{n+j} = v_j$; hence $\sigma^n x \in V$. $\square$

Lemma 6 (Dimension groups). For $n \neq 0$, $R([n]) = \mathbb{Q}$, $\mathcal{G}_{[n]} = \{q : qn^k \in \mathbb{Z} \text{ for some } k\}$ and $\mathcal{G}_{[n]}^+ = \{q : qn^k \in \mathbb{Z}_{\geq 0} \text{ for some } k\}$. Moreover $\mathcal{G}_{[2]} = \mathcal{G}_{[4]}$ and $\mathcal{G}_{[2]}^+ = \mathcal{G}_{[4]}^+$, so the identity is an order isomorphism. (Both equal $\mathbb{Z}[1/2]$ with its usual cone.)

Proof. $x[n]^k = xn^k$ and $x = (x/n)[n]$. If $q2^k \in \mathbb{Z}$ then $q4^k = (q2^k)2^k \in \mathbb{Z}$; if $q4^k \in \mathbb{Z}$ then $q2^{2k} \in \mathbb{Z}$. The same holds with $\mathbb{Z}_{\geq 0}$. $\square$

Lemma 7 (Fixed points). *Let $\varphi : X \rightarrow Y$ be a bijection between subshifts with $\varphi(\text{orb}(x)) \subseteq \text{orb}(\varphi(x))$ for all x . If $\sigma y = y$, then $\sigma(\varphi^{-1}y) = \varphi^{-1}y$.*

Proof. A fixed configuration is constant (induction on $\mathbb{Z}$), so $\sigma^k y = y$ for all $k \in \mathbb{Z}$ and $\text{orb}(y) = \{y\}$. Put $x = \varphi^{-1}y$. Since $\sigma x \in \text{orb}(x)$, $\varphi(\sigma x) \in \text{orb}(y) = \{y\} = \{\varphi(x)\}$, so $\sigma x = x$ by injectivity. $\square$

Theorem 8. *For $m < n$ there is no bijection $X_{[m]} \rightarrow X_{[n]}$ mapping every orbit into an orbit. In particular $X_{[2]}$ and $X_{[4]}$ are not orbit equivalent.*

Proof. The n constant sequences c_e (e an edge of $[n]$) are fixed points of $X_{[n]}$. By the lemma, each $\varphi^{-1}(c_e)$ is a fixed, hence constant, point of $X_{[m]}$, determined by its value at 0. So $e \mapsto \varphi^{-1}(c_e)_0$ is an injection from an n -set into an m -set, contradicting $m < n$. $\square$

Theorem 9. *$X_{[2]}$ and $X_{[4]}$ are mixing SFTs with order-isomorphic dimension groups that are not orbit equivalent. Hence the “if” direction of the first clause fails, and the conjecture is false.*

4 Lean formalization and axiom audit

File `lean/Conjecture9629/Basic.lean` (namespace `C9629`), Lean 4 with Mathlib, only `import Mathlib`. Main theorems:

- `counterexample : IsSFT (edgeShift (full 2))  $\wedge$  IsSFT (edgeShift (full 4))  $\wedge$  IsMixing (edgeShift (full 2))  $\wedge$  IsMixing (edgeShift (full 4))  $\wedge$  DimOrderIso (full 2) (full 4)  $\wedge$   $\neg$  OrbitEquivalent (edgeShift (full 2)) (edgeShift (full 4)).`
- `not_OEDimCriterion :  $\neg$  OEDimCriterion`, where `OEDimCriterion` says: for all r, s and all A ($r \times r$), B ($s \times s$) over $\mathbb{N}$, if X_A and X_B are mixing then `OrbitEquivalent  $X_A$   $X_B$   $\iff$  DimOrderIso A B`.
- `conjecture_false (entropyClause krClause : Prop) :  $\neg$  (OEDimCriterion  $\wedge$  entropyClause  $\wedge$  krClause).`
- Supporting: `no_orbit_map` (no bijection $X_{[m]} \rightarrow X_{[n]}$, $m < n$, maps orbits into orbits), `symm_fixed`, `isMixing_of_eq_univ`, `mem_dimGroup_full`, `mem_dimCone_full`, `dimOrderIso_two_four`.

Conventions: matrices act on row vectors (`vecMul`); $\mathbb{N}$ contains 0 and $k \geq 0$ in the definition of $\mathcal{G}_A$; Mathlib’s `shift k x` is $i \mapsto x(k + i)$.

Axiom audit. `#print` axioms for `counterexample`, `not_OEDimCriterion`, `conjecture_false` and `no_orbit_map` reports exactly `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom. Checked with `lake build` (Lean v4.33.1, Mathlib v4.33.1). The Lean file has 381 lines.

Sources

- [S] S. Schmieding, *Automorphisms of the shift: Lyapunov exponents, entropy, and the dimension representation*, arXiv:1803.04060, Sec. 2: “Let A be a $k \times k$ square matrix over $\mathbb{Z}_+$. We let $R(A)$ denote the eventual range subspace $\mathbb{Q}^k A^k \subset \mathbb{Q}^k$ (throughout we will always assume matrices to act on row vectors). The dimension triple $(\mathcal{G}_A, \mathcal{G}_A^+, \delta_A)$ associated to A consists of the abelian group $\mathcal{G}_A$, the semi-group $\mathcal{G}_A^+ \subset \mathcal{G}_A$, and the automorphism δ_A of $\mathcal{G}_A$, where (1) $\mathcal{G}_A = \{x \in R(A) \mid xA^k \in \mathbb{Z}^k \text{ for some } k \geq 0\}$. (2) $\mathcal{G}_A^+ = \{x \in R(A) \mid xA^k \in (\mathbb{Z}_+)^k \text{ for some } k \geq 0\}$. (3) $\delta_A(x) = xA$. While the definition of $(\mathcal{G}_A, \mathcal{G}_A^+, \delta_A)$ above relies on the matrix A , there is an alternative definition, due to Krieger, which is built more directly from the system (X_A, σ_A) .”
<https://arxiv.org/abs/1803.04060>

- [E] J. Epperlein, *Eventual conjugacy of free inert G -SFTs*, arXiv:2309.08512: “Every essential graph Γ defines an edge shift X_Γ . This subshift of finite type over the alphabet $E(\Gamma)$ consists of all biinfinite sequences of edges $(e_k)_{k \in \mathbb{Z}}$ with $i(e_{k+1}) = t(e_k)$ for all $k \in \mathbb{Z}$. [...] For $A \in \mathbb{Z}_+^{V \times V}$ we can canonically label the edges in the graph defined by A from i to j by $(i, j, 1), (i, j, 2), \dots, (i, j, A_{i,j})$ ”; and “The dimension group D_A of X_A is defined as the direct limit of the diagram $\mathbb{Z}^V \xrightarrow{A} \mathbb{Z}^V \xrightarrow{A} \dots$ [...] One can further endow this group by an order and the automorphism induced by the shift map to obtain the so called dimension triple”.
<https://arxiv.org/abs/2309.08512>

- [W1] Wikipedia, *Subshift of finite type*: “every subshift of finite type is topologically conjugate to an edge shift via a local recoding.” https://en.wikipedia.org/wiki/Subshift_of_finite_type

- [W2] Wikipedia, *Mixing (mathematics)*: “A system is said to be topologically mixing if, given open sets A and B , there exists an integer N , such that, for all $n > N$, one has $f^n(A) \cap B \neq \emptyset$.”
[https://en.wikipedia.org/wiki/Mixing_\(mathematics\)](https://en.wikipedia.org/wiki/Mixing_(mathematics))